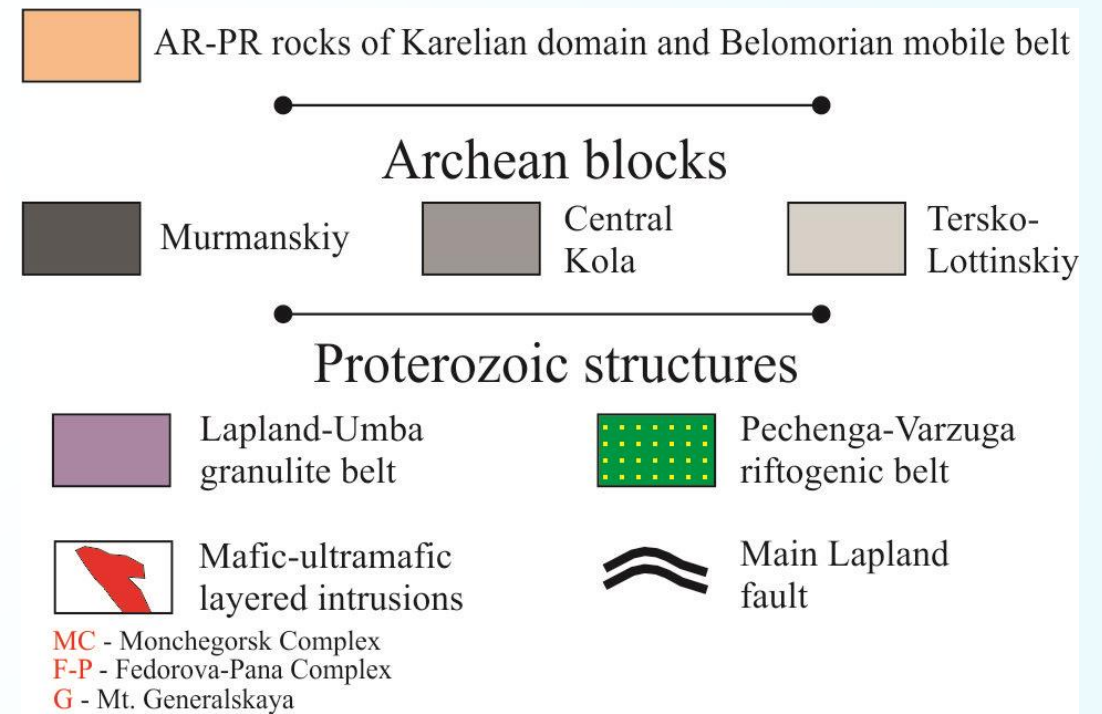
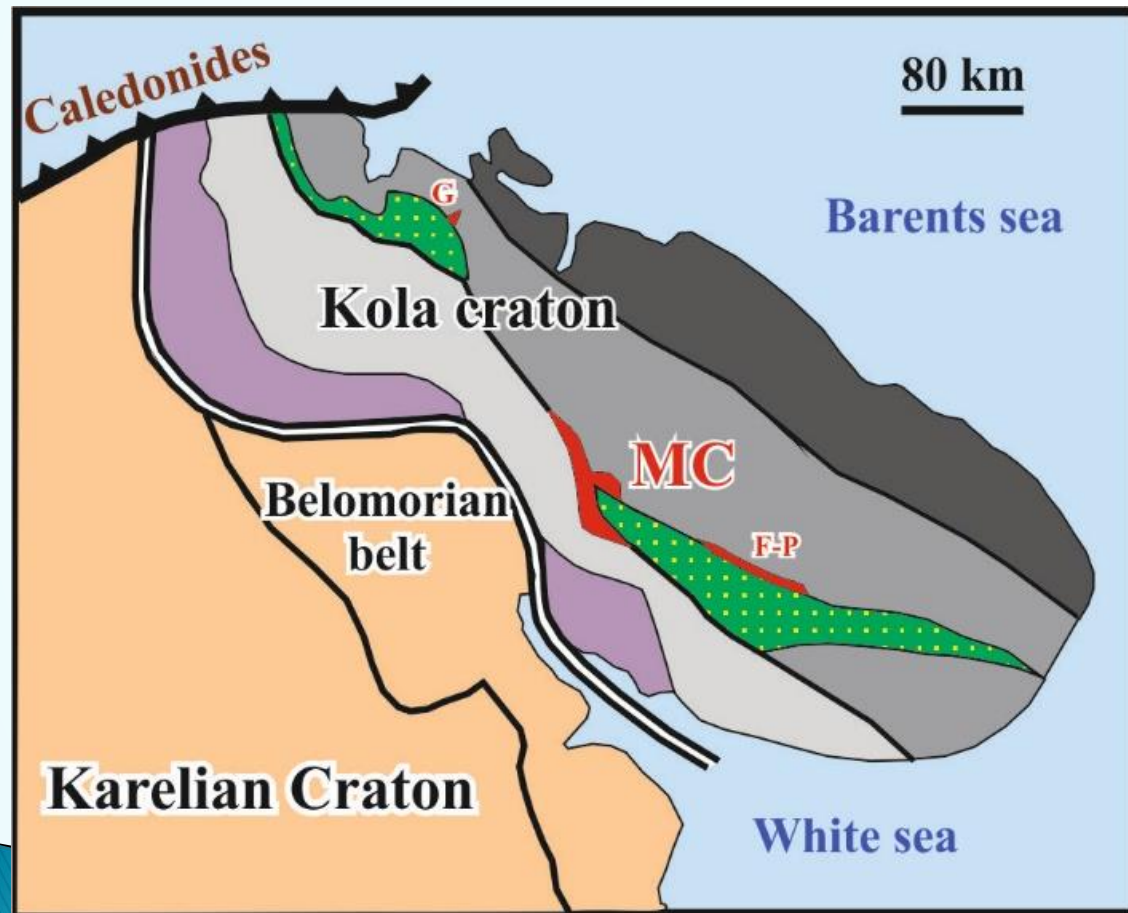


The role of metasomatism in PGE-bearing rocks formation of the Vuruchuaivench massif, Monchegorsk Complex (Kola Peninsula, Russia)

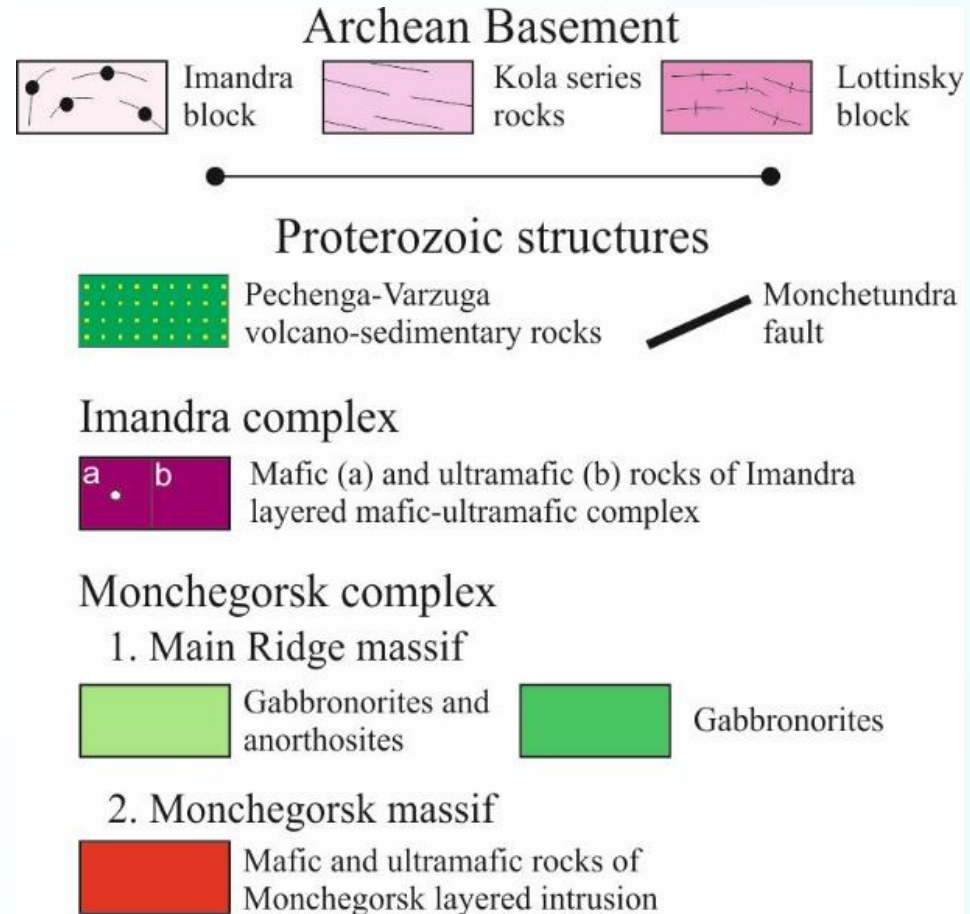
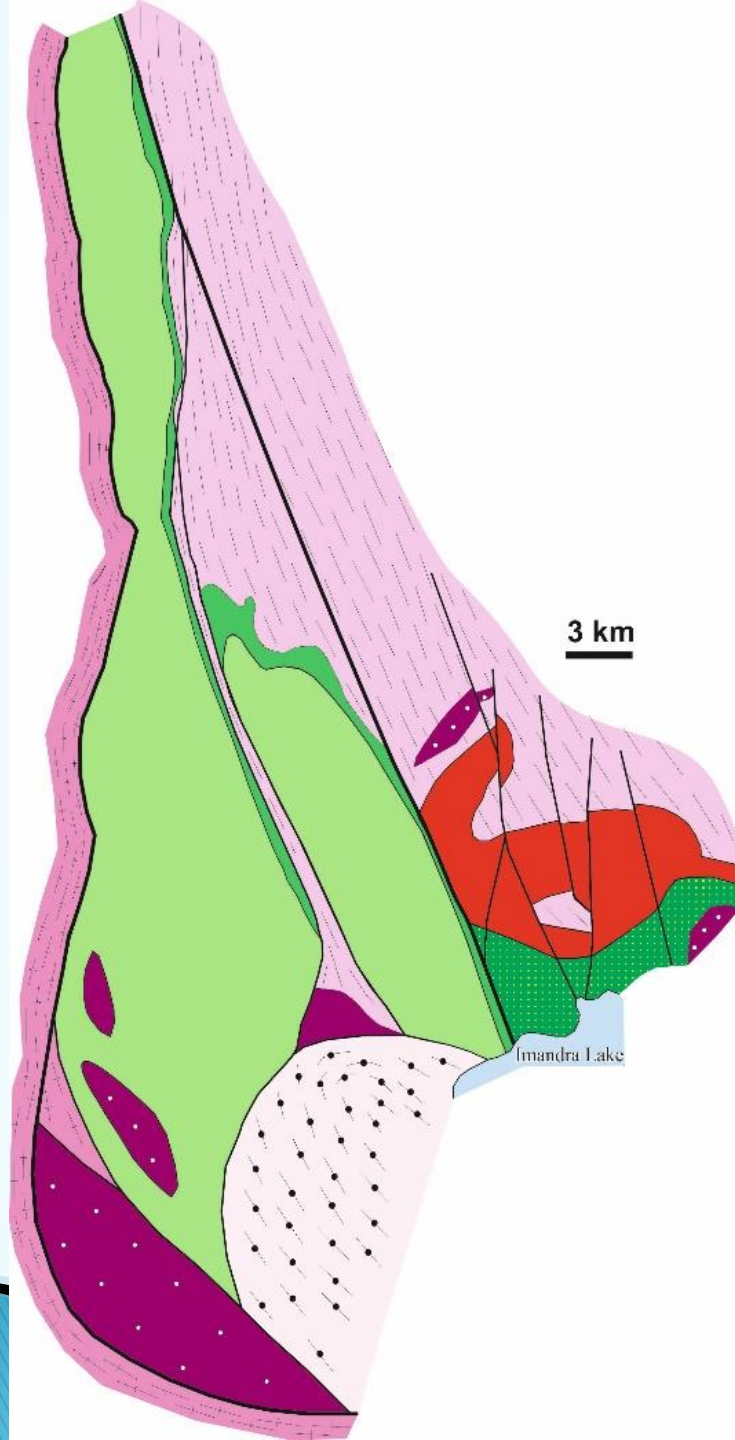
ALEKSEI PAKHALKO and KASHIN SERGEY

A.P. Karpinsky Russian Geological Research Institute (VSEGEI), Sredniy prospect 74, Saint-Petersburg, Russia, Aleksey_Pahalko@vsegei.ru

Position of layered complexes in main tectonic structures of north-western part of Baltic shield



Schematic geological map of Monchegorsk complex

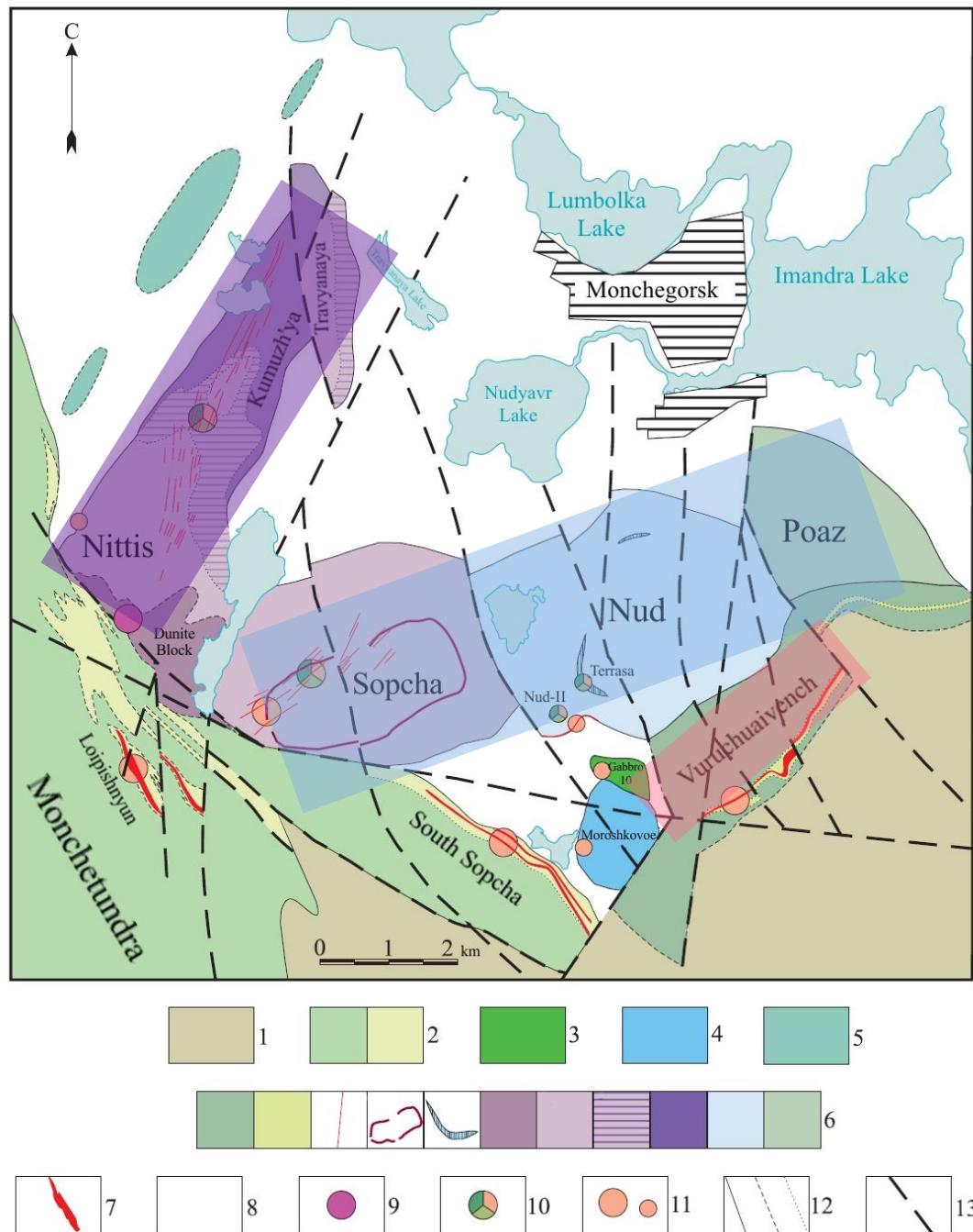


Modified from *Sharkov, E.V., and Chistyakov, A.V., 2012*

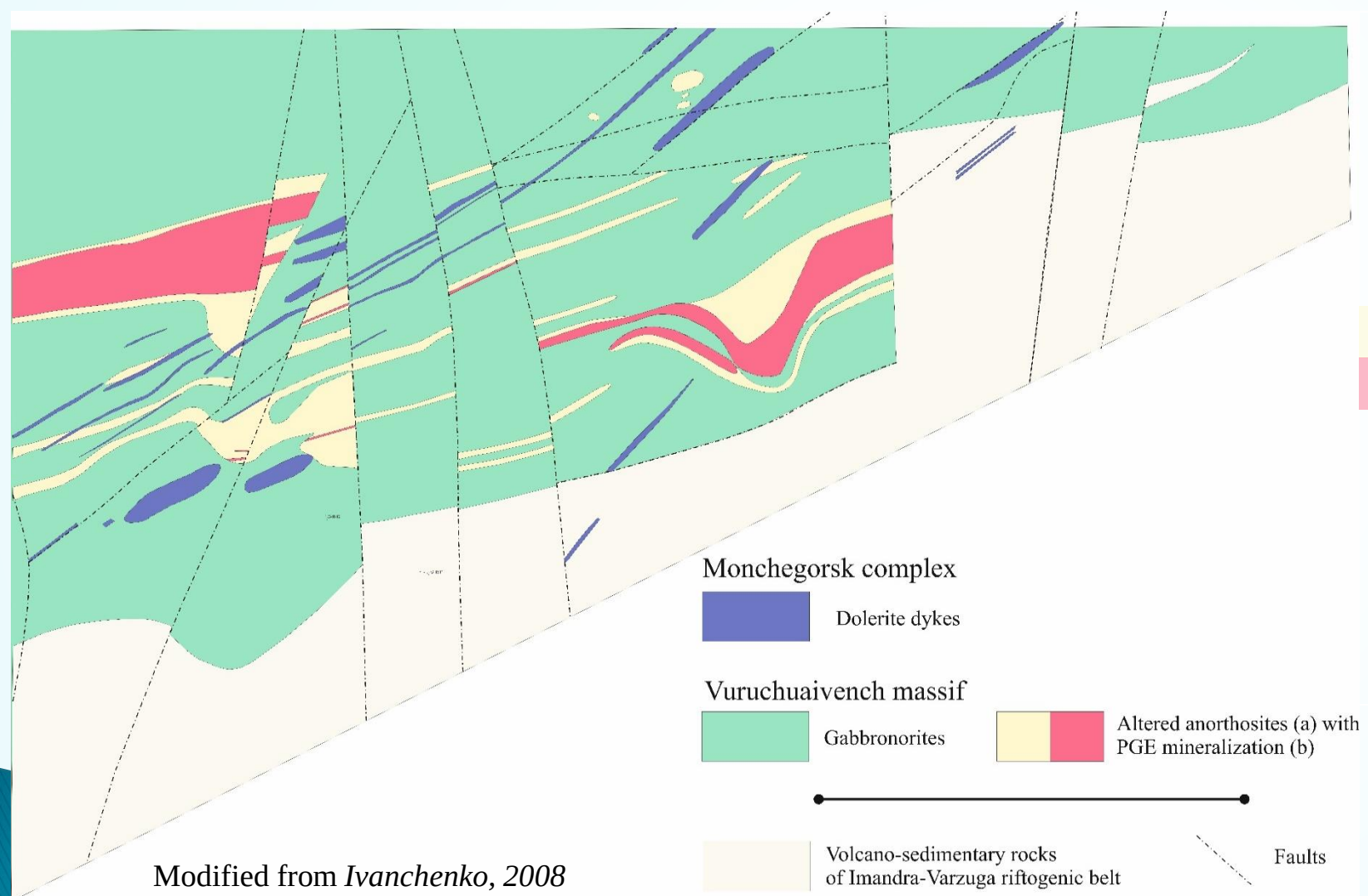
Geology of Monchegorsk massif (Monchepluton)

The Monchegorsk massif belongs to the complex and consists of two branches: the sub-longitude peridotite-pyroxenite trend (Nittis, Kumuzh'ya, and Travyanaya (NKT) massifs), and the sub-latitude pyroxenite-gabbro-norite trend (Sopcha, Nyud, and Poaz massifs).

The Vuruchuaivench adjoins the Nyud and Poaz massifs from the southeast, and contact between them is tectonic.



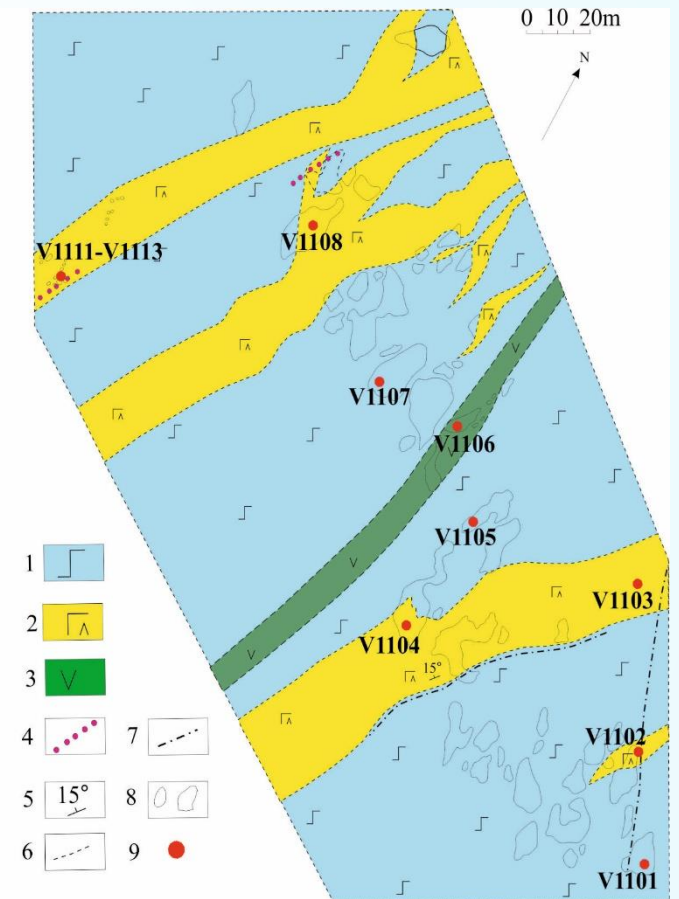
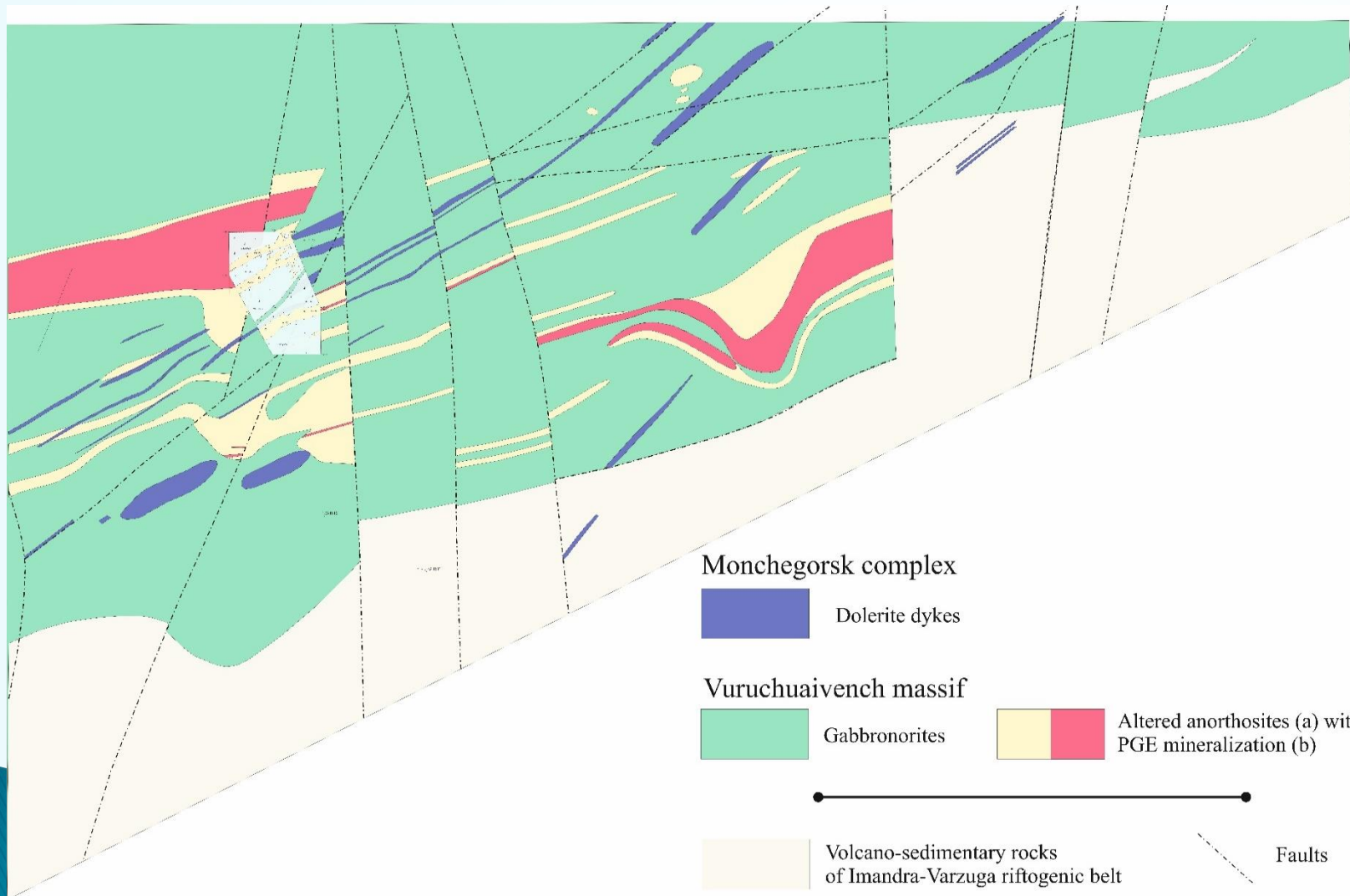
Geology of Vuruchuaivench massif



Modified from *Ivanchenko, 2008*

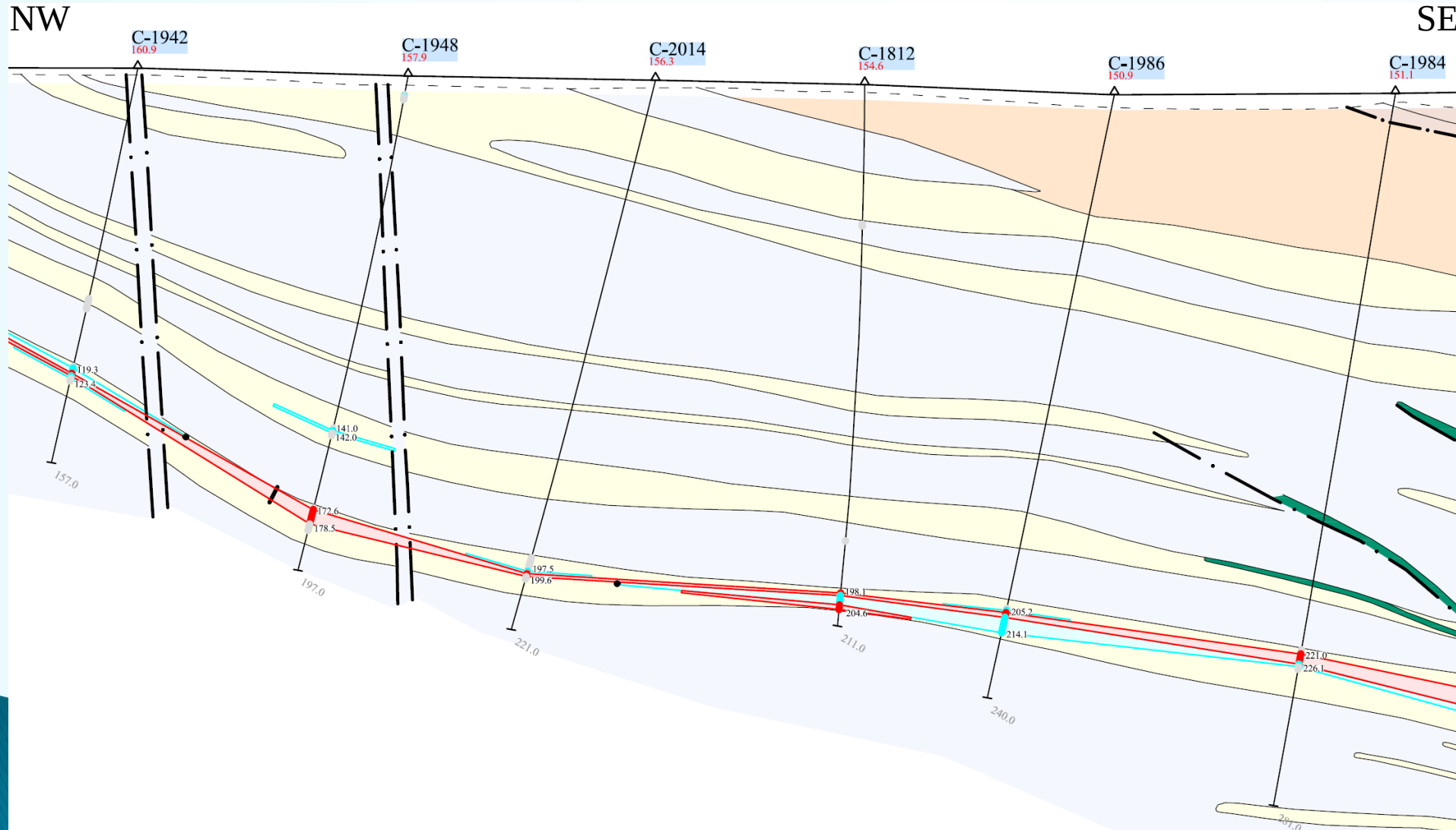
The **Vuruchuaivench** (also known as Vurechuaivench) massif supposedly is the **upper layered unit of the Monchegorsk pluton**. The Vuruchuaivench massif consists of altered massive gabbronorites and anorthosites. The PGE-bearing lens-like bodies are mostly related to anorthosites but occasionally can be found in gabbronorites. Ore bodies are always associated with up to several wt.% of sulfides. Palladium concentrations (4 ppm on average) dominate over Pt (0.5 ppm on average)

Geology of Vuruchuaivench massif



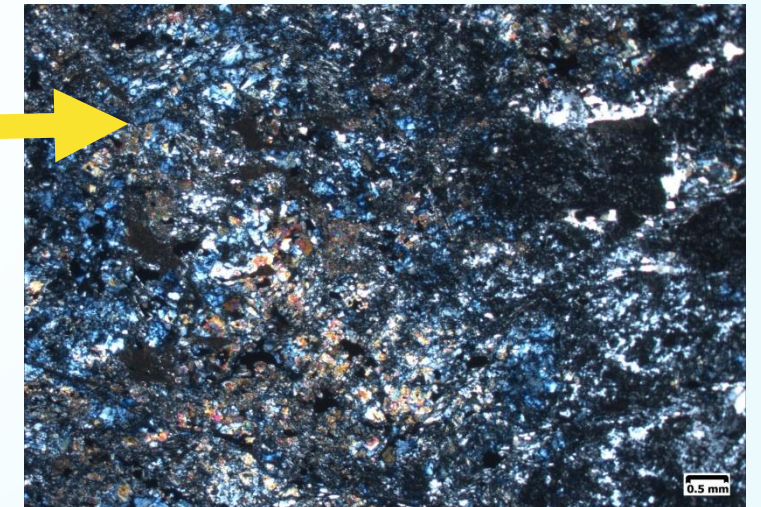
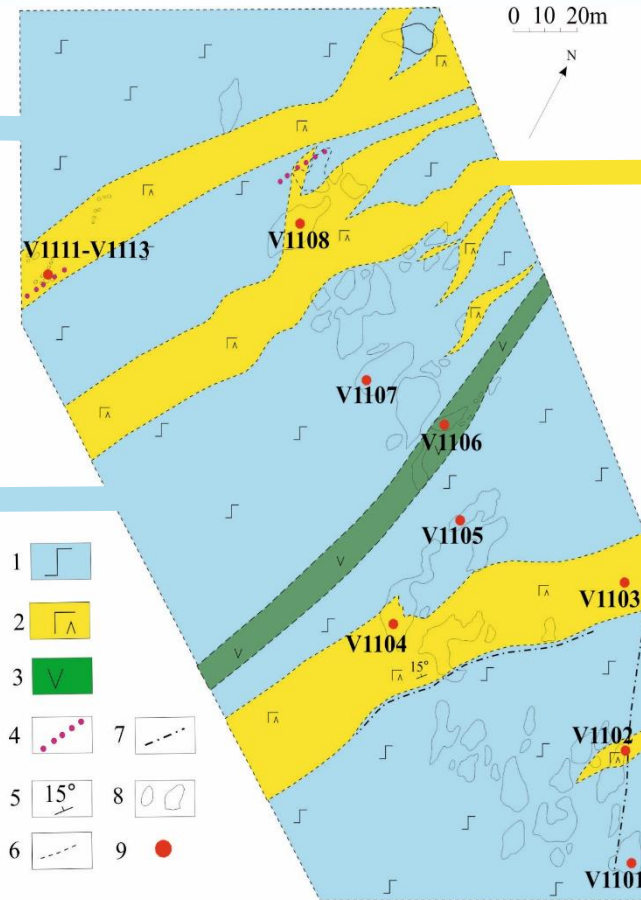
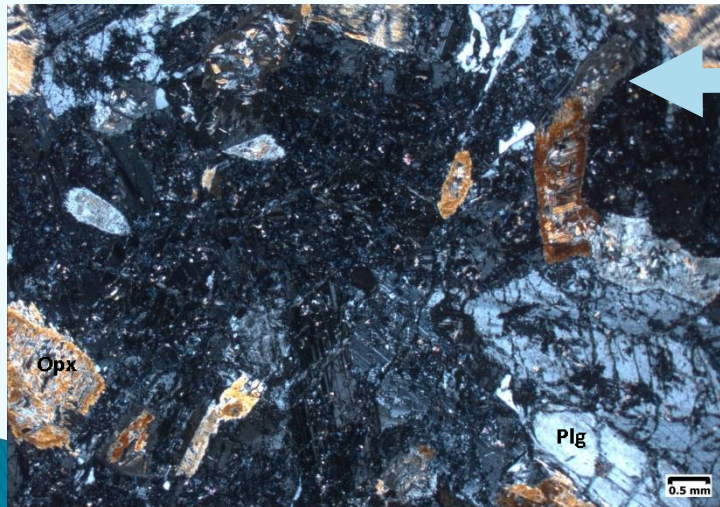
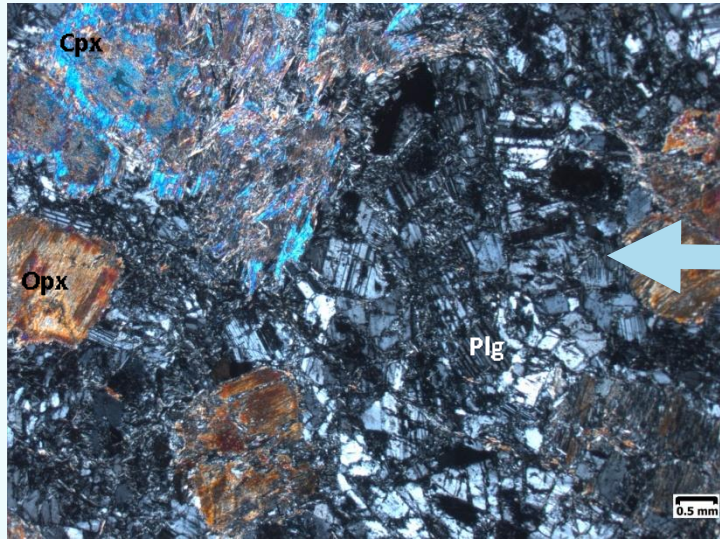
Modified from Pripachkin et.al., 2013

Geology of Vuruchuaivench massif



Alternation of
gabbro-norites and
anorthosites
Not every
anorthosite is ore-
bearing

Mineral composition of rocks



“Gabbronorites”

“Anorthosites”

Chemical composition of rocks

Component %	“Gabbro norites”		“Anorthosites”				
	V1101	V1105	V1103	V1104	V1111	V1112	V1113
			Non-ore		Ore-bearing area		Non-ore
SiO ₂	51,5	50,6	51,3	51,8	46,6	48,4	49,6
Al ₂ O ₃	20,3	19,3	25,5	23,8	24,1	24,3	26,3
TiO ₂	0,24	0,22	0,21	0,23	0,2	0,35	0,35
ΣFe	5,28	6,2	2,95	3,11	6,66	4,91	2,78
MnO	0,095	0,11	0,046	0,062	0,076	0,066	0,059
MgO	6,53	8,94	2,57	3,69	4,53	3,62	0,81
CaO	11,5	9,68	10,2	11,7	12,5	12,4	14,9
Na ₂ O	2,4	1,83	4,84	3,48	2,26	3,05	3,61
K ₂ O	0,22	0,19	0,53	0,83	0,088	0,16	0,25
P ₂ O ₅	0,025	0,025	0,025	0,025	0,025	0,025	0,025
l.o.i	1,85	2,71	1,82	1,37	2,86	2,61	1,39
Sum	100	99,8	100	100	99,8	99,9	100
S	0,0093	0,0025	0,0025	0,0025	0,38	0,17	0,0025
Co	31,1	42,5	13,2	14,3	65,8	31,1	5,64
Ni	101	144	38,9	51,3	1660	471	59,4
Cu	26,8	13,1	14	16,6	1300	676	57,5
Zn	44,7	44,5	21	22,6	34,9	33,7	12,8
Pd	0,001	0,001	0,001	0,0064	3,95	1,86	0,032
Pt	0,0046	0,0049	0,0052	0,0028	0,59	0,27	0,045
ΣPGE	0,0087	0,0093	0,0072	0,0113	4,64	2,174	0,078

Mineral	“Gabbro norites”		“Anorthosites”			
	V1101	V1105	V1103	V1104	V1111	V1112
Quartz	1,75	3,38	0	0	0	0
Plagioclase	70,16	65,22	83,84	80,13	81,29	83,99
Orthoclase	1,55	1,35	3,53	5,61	0,62	1,12
Nepheline	0	0	4,15	0	0	0
Diopside	9,42	2,95	2,98	7,79	4,8	6,67
Hypersthene	16,54	26,51	0	2,02	4,83	0,76
Olivine	0	0	5,08	4	7,41	6,56
Ilmenite	0,28	0,26	0,24	0,26	0,24	0,41
Magnetite	0,21	0,26	0,12	0,13	0,27	0,2
Pyrite	0,01	0	0	0	0,48	0,21
Total	99,99	100	100	100,01	100,01	99,99

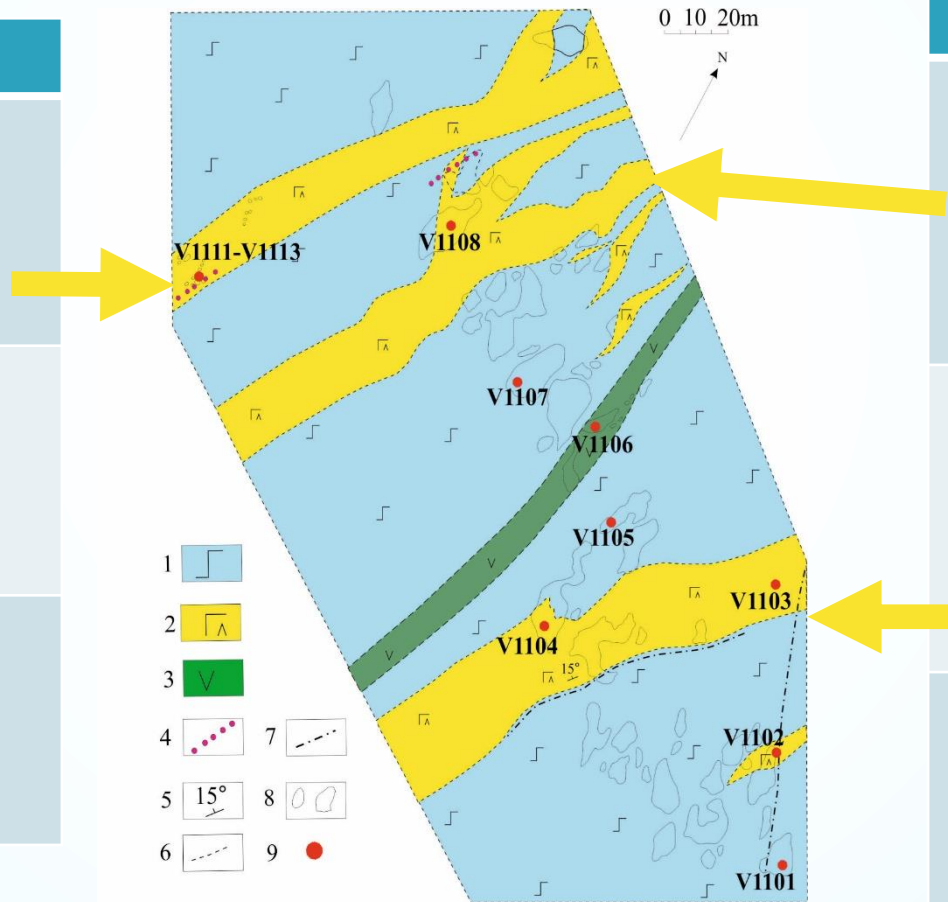
Normative mineral calculation

Johannsen, A., 1931, A Descriptive Petrography of the Igneous Rocks, Volume 1 (p. 88-92). University of Chicago Press, Chicago, 267 p

X-ray analysis of rocks

Ore-bearing area

Sample	Minerals, %
V1111	Clinochlore – 36 Quartz – 30 Albite – 20 Zoisite – 9
V1112	Quartz – 33 Albite – 30 Clinochlore – 26 Zoisite – 10
V1100	Albite – 37 Clinochlore – 20 Sericite – 20 Quartz – 15



Non-mineralized area

Sample	Minerals, %
V1108	Albite – 40 Zoisite – 26 Hornblend – 6 Quartz – 10 Clinochlore – 13
V1103	Albite – 50 Clinochlore – 12 Zoisite – 12 Sericite – 10 Quartz – 5
V1104	Albite – 56 Hornblend – 12 Quartz – 9 Clinochlore – 4

Alteration of Vuruchuaivench massif's rocks

The “Vurechuaivench” prospect is confined to the zone of alternating meso-leucocratic massive and taxitic gabbronorite and anorthosite-plagioclase. The PGE mineralization is related to the extensive zone of lens-like and tabular sulphide bodies. The

Pripachkin et. al. 2013 **Overview**

which experienced significant crustal contamination. The results of Sm–Nd isotopic geochronological study of ore-bearing metaplagioclase from PGE reef of the Vurechuaivench deposit (2410 ± 58 Ma at $\epsilon_{Nd} = -2.4$) provide evidence for the appreciable effect of metamorphic and hydrothermal metasomatic alterations on PGE ore formation. The Sm–Nd

Chashchin et. al. 2016 **Sm–Nd isotopy**

Poaz massifs. Mineral compositions of gabbronorites of the Vuruchuaivench and Nyud–Poaz massifs are almost identical. However, there are some specific differences: gabbronorites of the piedmont have a more leucocratic composition, and are characterized by an elevated amount of monoclinic pyroxene and extensive development of autometamorphism (Kozlov *et al.*, 1967). These differences can be related to the position

Grokhovskaya et. al. 2000

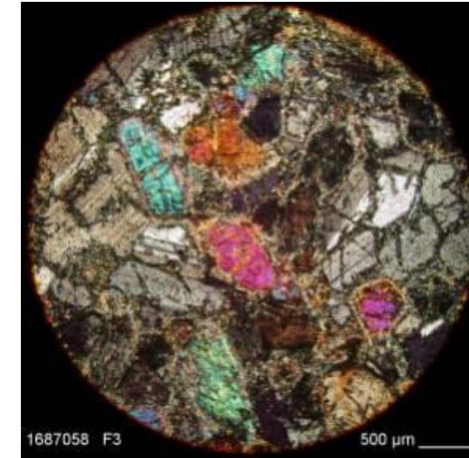
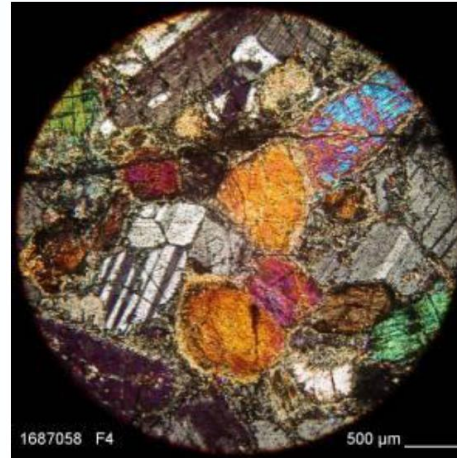
Mineralogy of PGM

Vuruchuaivench intrusion: The Vuruchuaivench intrusion consists of medium-grained leucogabbronorite and minor anorthosite, both of which are pervasively altered; however, primary textures were preserved. The intrusion is believed to

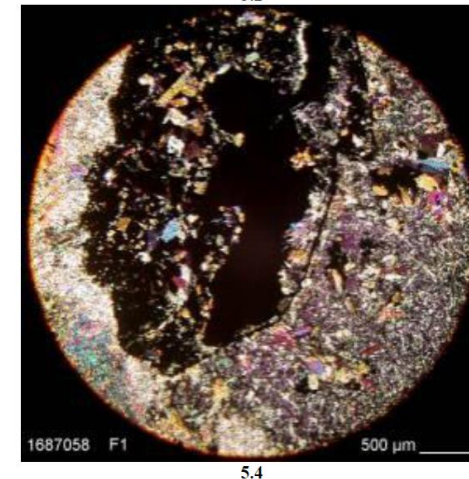
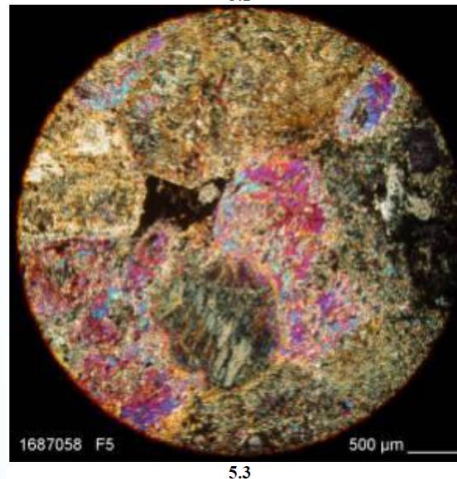
Karykowski et. al. 2018 **Petrology, PGE mineralization origin**

Alteration of Vuruchuaivench massif's rocks

The Vuruchuaivench prospecting area located on upper part of layered section of Monchegorsk massif (~2.45 GA) which comprise the gabbronorites, leucogabbronorites and plagioclasicites (fig. 1). The PGE and gold content in ore horizon (n*0.1 – up to 17m) ranged in 2 – 6 ppm, Pd : Pt = ~8 : 1, Cu-Ni sulfides – up to n*1%. The host rocks, in relation to ore horizon, are taxitic and pegmatiodic varieties of gabbronorites and leucogabbronorites, which are strongly altered and the primary rock-forming minerals are replaced by secondary hydroxyl-bearing minerals.

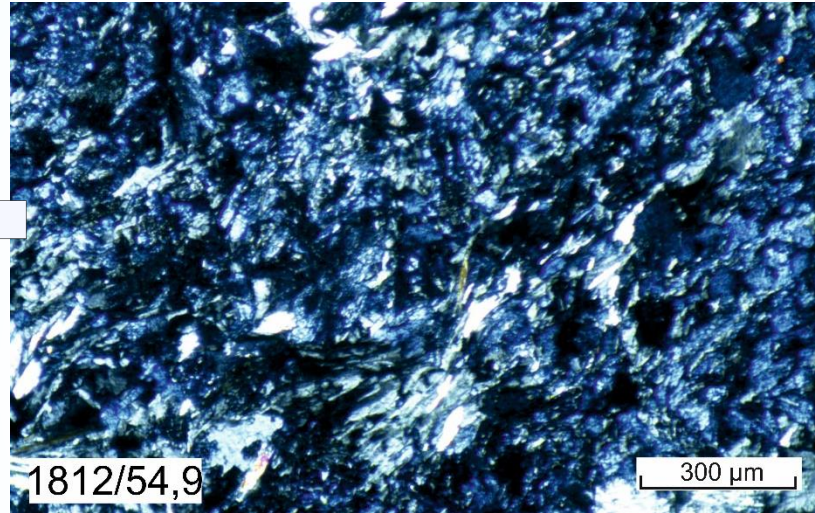
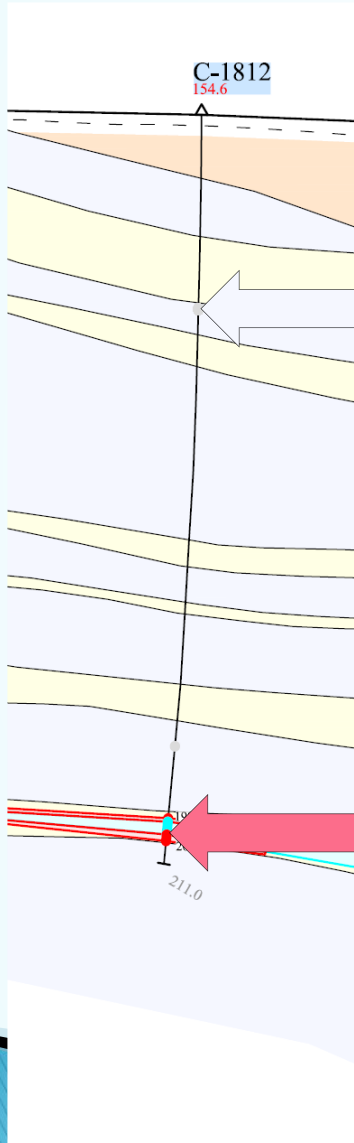


Altered rocks

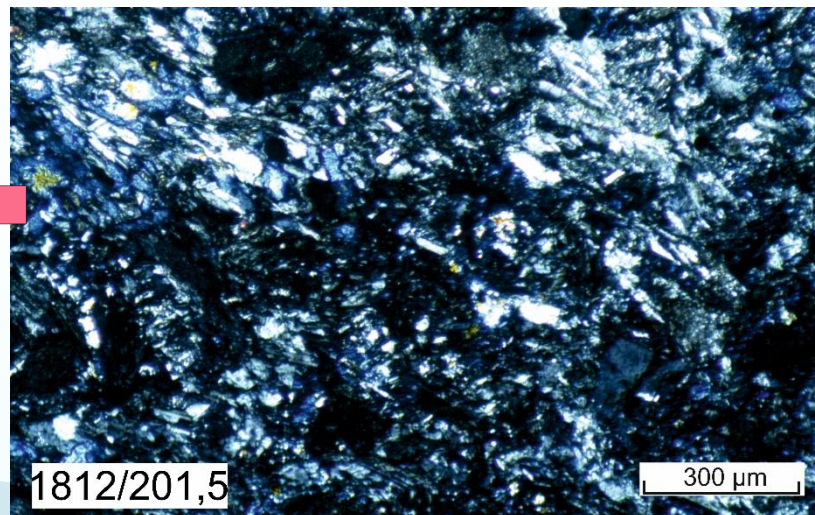


Fully altered rocks
**Relicts of magmatic?
structure**

Alteration of Vuruchuaivench massif's rocks

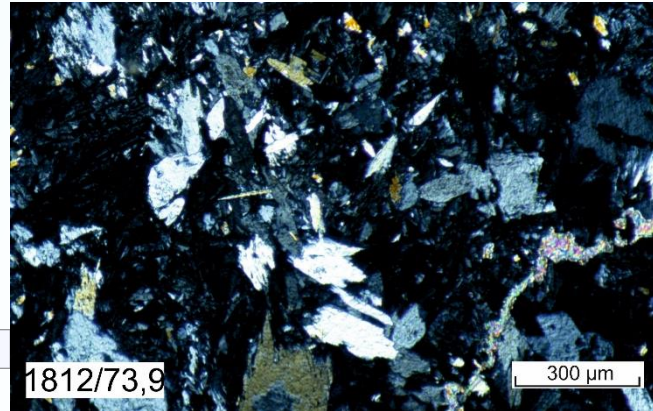
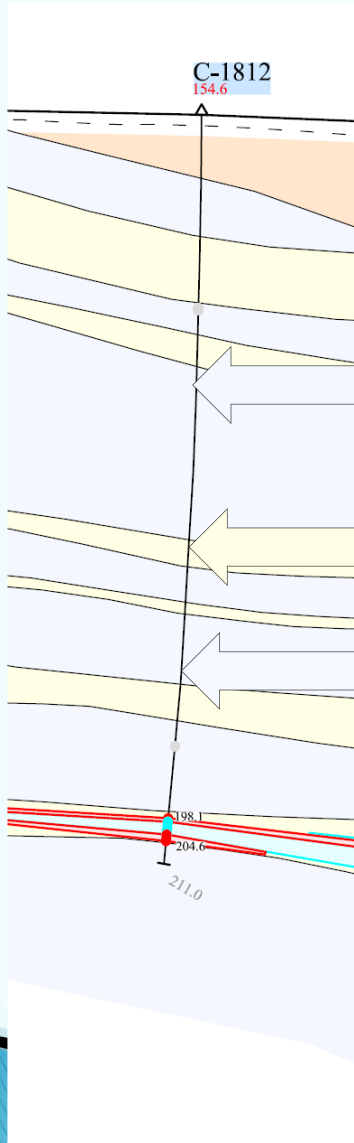


Pervasively metasomatized ore-bearing ($\Sigma\text{PGE}=0.63$ ppm) rock within “gabbronorites”

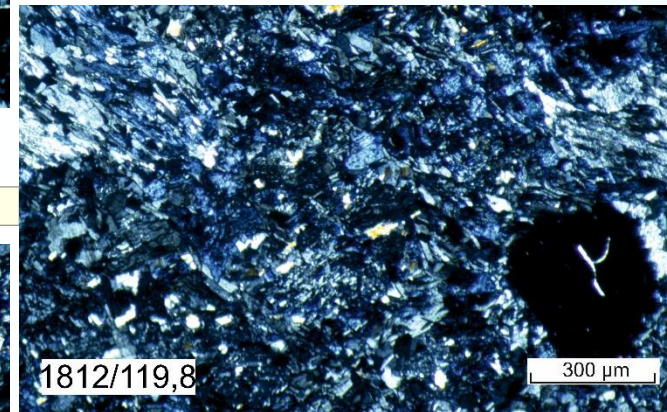


Pervasively metasomatized ore-bearing ($\Sigma\text{PGE}=0.75$ ppm) rock within “anorthosites”

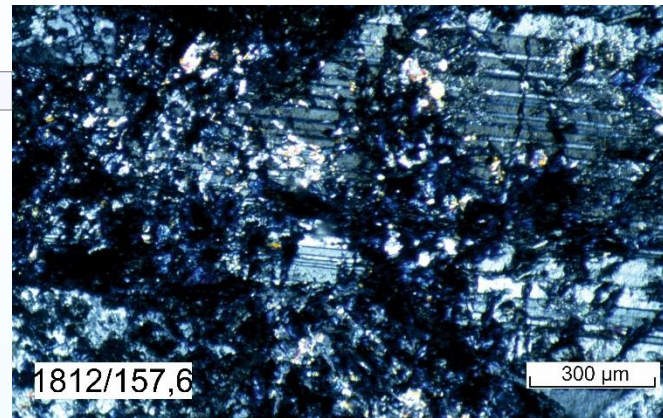
Alteration of Vuruchuaivench massif's rocks



**Medium-fine-grained
metasomatic Q-Alb-Chl rock**

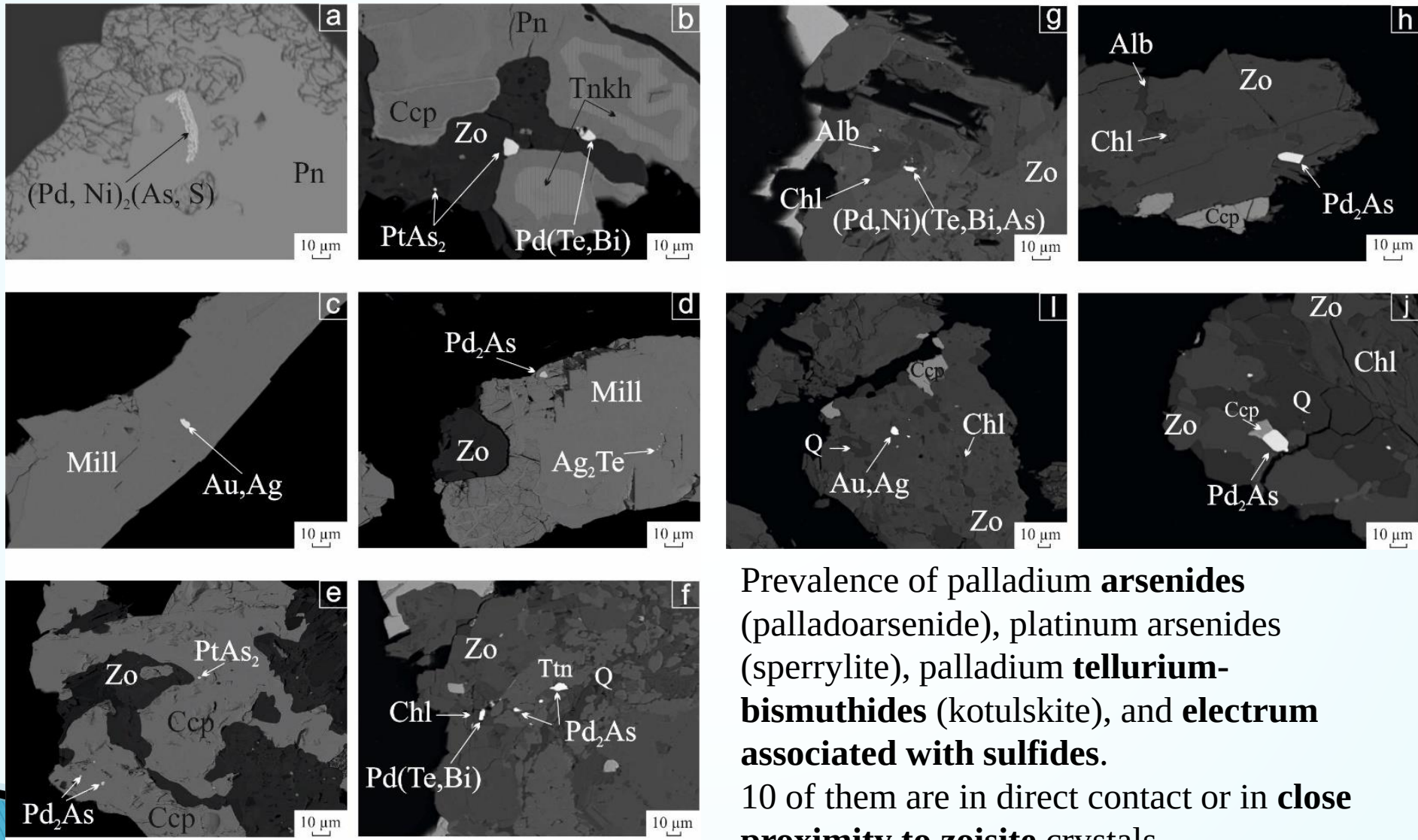


**Fine-grained
metasomatic Q-
Alb—Zo-Chl
rock**



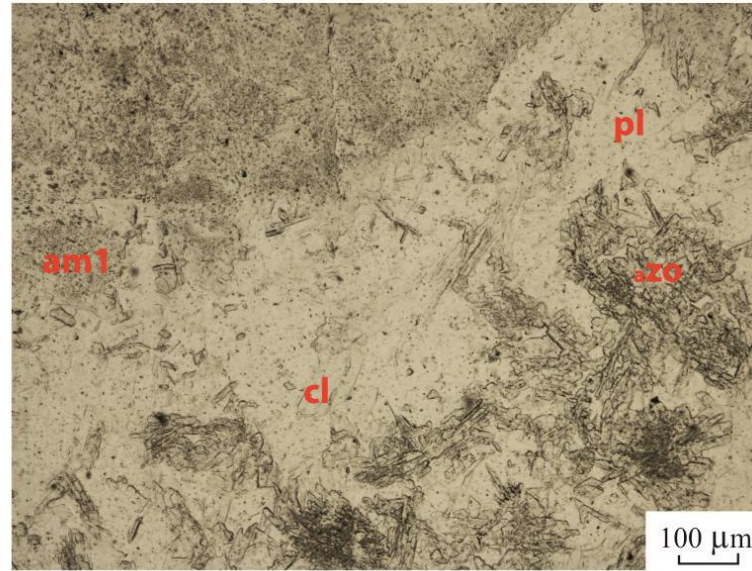
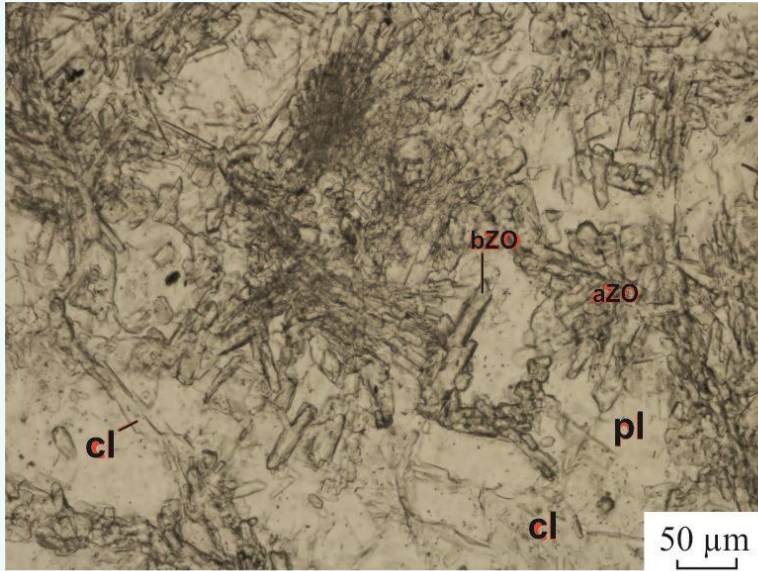
**Metasomatic Alb-Q-Zo-Chl
rock with Plg relicts**

Platinum group elements mineralization



Prevalence of palladium **arsenides** (palladoarsenide), platinum arsenides (sperrylite), palladium **tellurium-bismuthides** (kotulskite), and **electrum** associated with sulfides. 10 of them are in direct contact or in **close proximity to zoisite** crystals

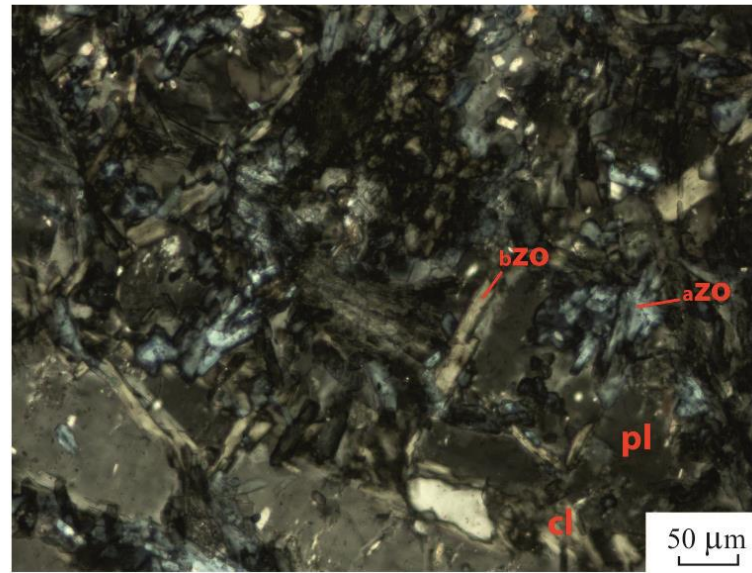
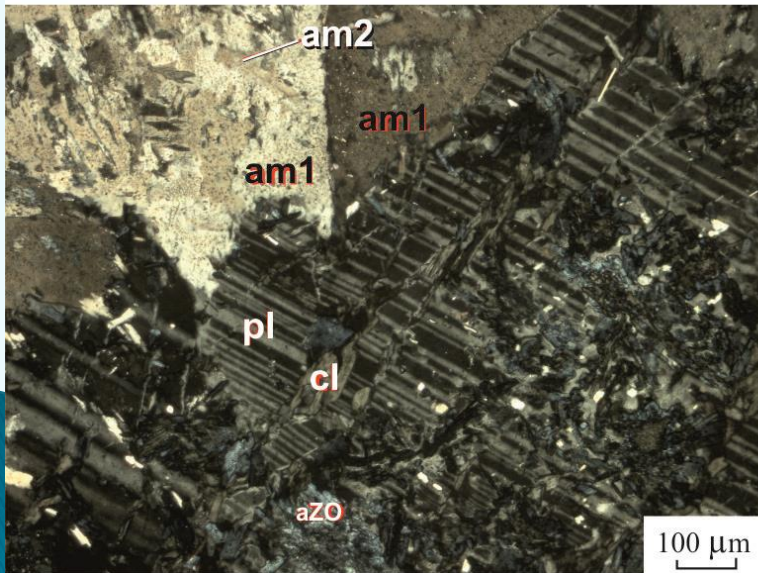
Additional mineralogical studies



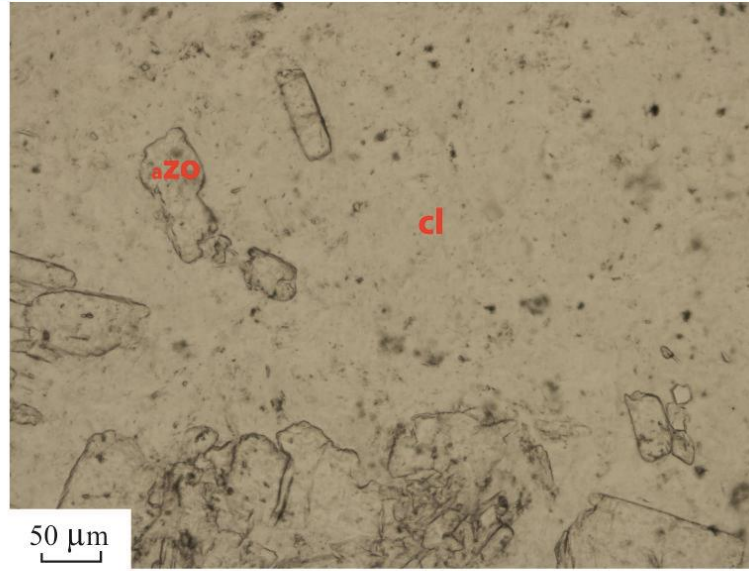
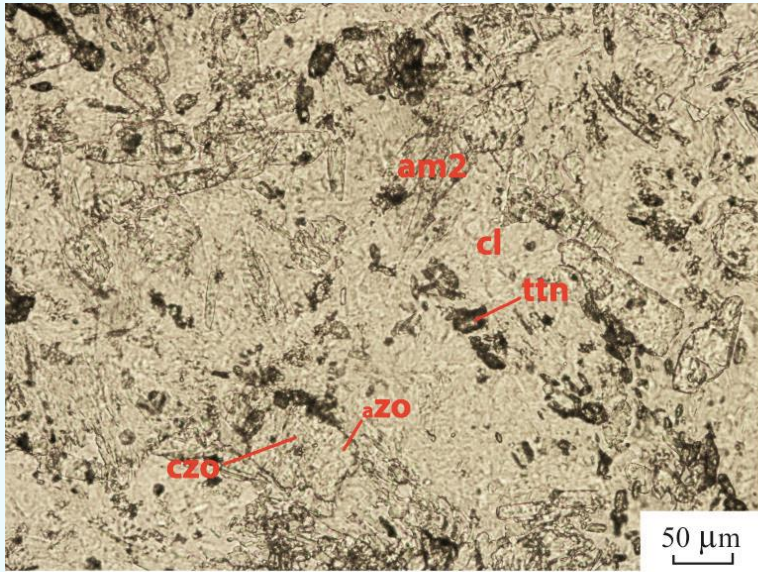
V1101

Non-mineralized area

- chlorite-zoisite propylitization with α and β -zoisite,
- two stages of amphibolization
- quartz-sericite (acidic) metasomatism

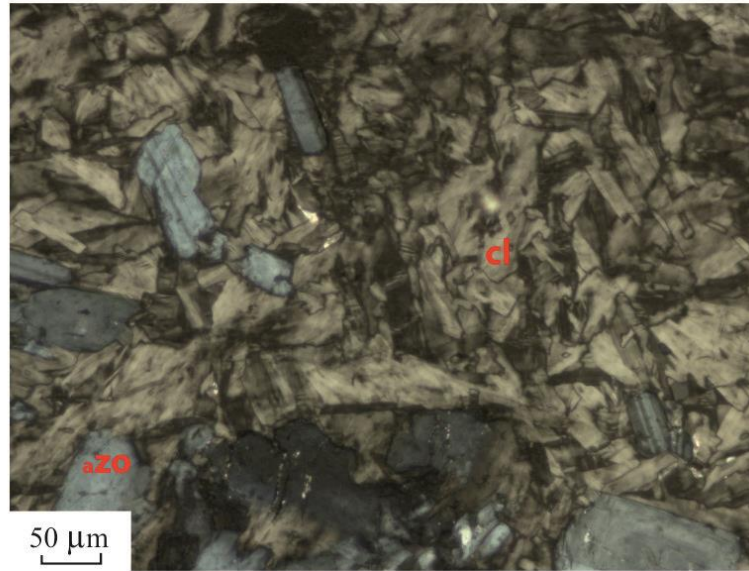
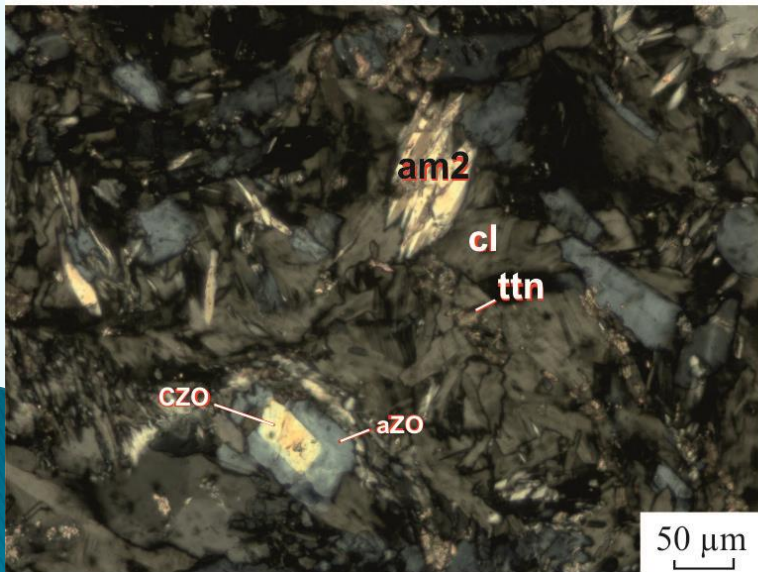


Additional mineralogical studies



1812/198 m.
Mineralized area

- chlorite-zoisite propylitization with α and β -zoisite,
- two stages of amphibolization
- quartz-sericite (acidic) metasomatism



Conclusion

- The rocks of the massif are significantly metasomatized, which is noted by many authors, but is often ignored.
- Ore-bearing rock should be pervasively altered but not every fully altered rock is PGE-mineralized
- Connection between metasomatism and the PGE mineralization is indicated. Association of ore minerals and primary magmatic not found.
- We established different stages of metasomatism which can help to reveal the full petrogenesis and development of PGE mineralization within Vuruchuaivench and similar to it massifs

Thank you for attention!



Sharkov, E.V., and Chistyakov, A.V., 2012, The early Paleoproterozoic Monchegorsk layered mafite-ultramafite massif in the Kola Peninsula: Geology, petrology, and ore potential: Petrology , v. 20, p. 607–639.